

IUPAC Rules	
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Types of alkyl groups:

CH ₃ - methyl
CH ₃ CH ₂ - (or) C ₂ H ₅ - ethyl

$\text{CH}_3\text{CH}_2\text{CH}_2-$	n-propyl (or) propyl
$\text{CH}_3-\underset{\text{CH}_3}{\underset{ }{\text{CH}}}-$	isopropyl (or) 1-methylethyl

$$\text{CH}_3$$

$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2-$	n-butyl (or) butyl
$\text{CH}_3-\underset{\text{ }}{\text{CH}}-\text{CH}_2-$	isobutyl (or) 2-methylpropyl

$$\text{CH}_3$$

$\begin{array}{c} \text{CH}_3\text{-CH}_2\text{-CH-} \\ \\ \text{CH}_3 \end{array}$	sec-butyl (or) 1-methylpropyl
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CH_3
|
 CH_3C tert butyl (or) 1,1 dimethylethyl

$$\begin{array}{c} \text{CH}_3\text{-C-} \\ | \\ \text{CH}_3 \end{array}$$
 tert-butyl (or) 1,1-dimethylethyl
$$\begin{array}{c} \text{CH}_3 \\ | \end{array}$$

CH₃-C - CH₂- neopentyl (or) 2,2-dimethylpropyl

CH₃

Among these prefixes **iso** and **neo** are considered for alphabetical order where as **n-**, **sec-** and **tert-** are not

considered for alphabetical order.

Abbreviations used for alkyl and aryl groups:
Me- Methyl, **Et-** Ethyl, **Pr-** Propyl, **Bu-** Butyl, **Ph-** Phenyl

Rules for IUPAC nomenclature of organic compounds:
IUPAC name of organic compound consist of three parts:

Prefix + Parent chain name (Word root) + Suffix

Word root: It indicates number of carbon atoms in the main chain

Suffix:

(1) **Primary suffix:** It is attached to word root to indicate whether the carbon chain is saturated or unsaturated. Eg. ane, ene and yne

(ii) Secondary suffix: It is added to primary suffix to indicate the functional group.
When adding the secondary suffix, the terminal 'e' is dropped if the secondary

suffix begins with a vowel and it is retained if the secondary suffix begins with a consonant.

Compounds with functional groups:

(1) **Alkenes:** Functional group is $>C=C<$ Suffix **-ene**

(i) $CH_2=CH_2$ Ethene (ii) $CH_3CH_2CH=CH_2$ But-1-ene

(iii) $CH_3CH=CHCH_3$ But-2-ene

(iv) $CH_2=CH-CH_2-CH=CH-CH_3$ Hexa-1,4-diene

(2) **Alkynes:** Functional group is $-C\equiv C-$ Suffix **-yne**

(i) $CH\equiv CH$ Ethyne (ii) $CH_3CH_2C\equiv CH$ But-1-yne (iii) $CH_3C\equiv CCH_3$ But-2-yne

(iv) $CH\equiv C-C\equiv C-C\equiv CH$ Hexa-1,3,5-triyne

(3) **Haloalkanes:** Functional group is halogen $-X$ (X- F, Cl, Br & I)

Prefix Halo- (fluoro, chloro, bromo, iodo)

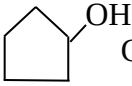
(i) CH_3Cl Chloromethane (ii) $CH_3CH(I)CH_3$ 2-Iodopropane

(iii) $CH_3CH(Br)CH_2CH_2Br$ 1,3-dibromobutane (iv) CH_2F-CH_2F 1,2-difluoroethane

(4) **Alcohols:** Functional group is $-OH$, hydroxyl group Suffix **-ol**

(i) CH_3OH Methanol (ii) $CH_3CH(OH)CH_3$ Propan-2-ol

(iii) $CH_2(OH)-CH_2(OH)$ Ethane-1,2-diol

(iv) $CH_2(OH)CH(OH)CH_2OH$ Propane-1,2,3-triol (v)  Cyclopentanol

(5) **Ethers:** Functional group is $-OR-$ Prefix for $-OR$ is alkoxy

CH_3O- methoxy, CH_3CH_2O- (or) C_2H_5O- ethoxy, $CH_3CH_2CH_2O-$ propoxy

(i) CH_3OCH_3 Methoxymethane (ii) $CH_3CH_2OCH_3$ Methoxyethane

(iii) $CH_3OCH_2CH_2CH_2OC_2H_5$ 1-Ethoxy-3-methoxypropane (alphabetical order)

(6) **Aldehydes:** Functional group is $\begin{array}{c} -C-H \\ || \\ O \end{array}$ (or) $-CHO$

Suffix **-al** (and) **-carbaldehyde** (Carbaldehyde is used when there are more than two $-CHO$ groups or if $-CHO$ is attached to cyclic hydrocarbon)

(i) $HCHO$ Methanal (ii) CH_3CHO Ethanal (iii) CH_3CH_2CHO Propanal

(iv) $OHC-CHO$ Ethane-1,2-dial (v) $OHCCH_2CH_2CH_2CHO$ Pentane-1,5-dial

(vi) $OHCCH_2CH(CHO)CH_2CHO$ Propane-1,2,3-tricarbaldehyde.

(7) **Ketones:** Functional group is $\begin{array}{c} \text{C-} \\ || \\ \text{O} \end{array}$ (or) -CO- Suffix **–one**

- (i) CH_3COCH_3 Propanone (ii) $\text{CH}_3\text{COCH}_2\text{CH}_3$ Butan-2-one
(iii) $\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$ Pentan-2-one (iv) $\text{CH}_3\text{COCH}_2\text{COCH}_3$ Pentane-2,4-dione

(6) **Carboxylic acid:** Functional group is $\begin{array}{c} \text{C-OH} \\ || \\ \text{O} \end{array}$ (or) -COOH

Suffix **–oic acid** (and) **–carboxylic acid** (for more than two -COOH groups or in cyclic)

- (i) HCOOH Methanoic acid (ii) CH_3COOH Ethanoic acid
(iii) $\text{CH}_3\text{CH}_2\text{COOH}$ Propanoic acid (iv) HOOC-COOH Ethane-1,2-dioic acid
(v) $\text{HOOCCH}_2\text{CH}_2\text{CH}_2\text{COOH}$ Pentane-1,5-dioic acid
(vi) $\text{HOOCCH}_2\text{CH}(\text{COOH})\text{CH}_2\text{COOH}$ Propane-1,2,3-tricarboxylic acid.

(7) **Esters:** Functional group is $\begin{array}{c} \text{C-OR} \\ || \\ \text{O} \end{array}$ (or) -COOR

Suffix **–oate** (and) **–carboxylate** (for more than two -COOR groups or in cyclic)

- (i) $\text{CH}_3\text{COOCH}_3$ Methyl ethanoate (ii) $\text{CH}_3\text{CH}_2\text{COOCH}_2\text{CH}_3$ Ethyl propanoate

(8) **Nitroalkanes:** Functional group is -NO_2 Prefix: **nitro**

- (i) CH_3NO_2 Nitromethane (ii) $\text{CH}_3\text{CH}(\text{NO}_2)\text{CH}_3$ 2-Nitropropane
(iii) $\text{CH}_2(\text{NO}_2)\text{-CH}_2\text{NO}_2$ 1,2-dinitroethane

(9) **Amines:** Functional group is -NH_2 , Suffix: **-amine**

- (i) CH_3NH_2 Methanamine (ii) $\text{CH}_3\text{CH}(\text{NH}_2)\text{CH}_3$ Propan-2-amine
(iii) $\text{CH}_2(\text{NH}_2)\text{-CH}_2(\text{NH}_2)$ Ethane-1,2-diamine

(10) **Amides:** Functional group is -CONH_2 ,

Suffix: **-amide** (and) **carboxamide** (for more than two -CONH_2 groups or in cyclic)

- (i) CH_3CONH_2 Ethanamide (ii) $\text{H}_2\text{NOCCH}_2\text{CH}_2\text{CONH}_2$ Butane-1,4-diamide

(11) **Nitriles or Cyanides:** Functional group is -CN ,

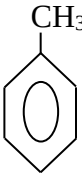
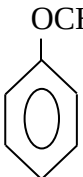
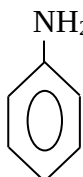
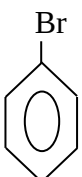
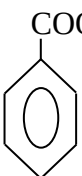
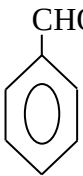
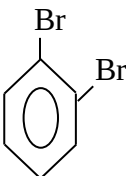
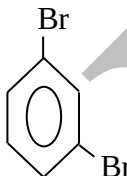
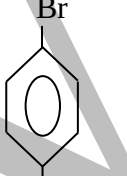
Suffix: **nitrile** (and) **carbonitrile** (for more than two -CN groups or in cyclic)

- (i) CH_3CN Ethanenitrile (ii) $\text{NCCH}_2\text{CH}_2\text{CN}$ Butane-1,4-dinitrile

(12) **Acid chlorides or Acyl chlorides:** Functional group is -COCl ,

Suffix: **oyl chloride** (and) **carbonylchloride** (for more than two -COCl groups or in cyclic)

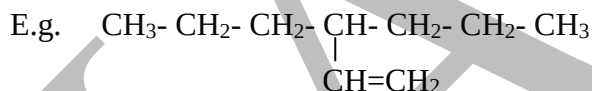
- (i) CH_3COCl Ethanoyl chloride (ii) $\text{C}_6\text{H}_5\text{COCl}$ Benzoyl chloride

Aromatic compounds			
			
Methylbenzene Toluene	Methoxy benzene Anisole	Benzenamine Aniline	Bromobenzene
			
Benzoic acid Benzenecarboxylic acid	Benzaldehyde Benzenecarbaldehyde	Phenol Benzenol	
			
o –dibromobenzene 1,2 –dibromobenzene	m –dibromobenzene 1,3 –dibromobenzene	p –dibromobenzene (Trivial names) 1,4 –dibromobenzene (IUPAC names)	
Ortho (1&2), meta (1&3) and para (1&4) represent two relative positions in benzene ring. These are used only in common names or trivial names.			
IUPAC names for multifunctional compounds:			
Order of priority of main (suffix) functional groups:			
Carboxylic acid (-COOH) > Esters (-COOR) > Acid halide (-COX) > Amides (-CONH ₂) > Nitriles (-CN) > Aldehydes (CHO) > Ketones (-CO=) > Alcohols (-OH) > Amines (-NH ₂) > Alkenes (C=C) > Alkynes (C≡C)			
Prefix functional groups:			
Halo (X), Nitro (NO ₂), Alkoxy (OR), Alkyl (R).			
For prefix functional groups the priority is based on alphabetical order			
Compound	Functional group	Prefix	Suffix
Halo compounds	-X X = F, Cl, Br, I	halo fluoro, chloro, bromo, iodo	
Nitro compounds	- NO ₂	nitro	
Ethers	- OR - OCH ₃	alkoxy methoxy	

	$-\text{OCH}_2\text{CH}_3$	ethoxy	
Alkenes	$>\text{C}=\text{C}<$		ene
Alkynes	$-\text{C}\equiv\text{C}-$		yne
Alcohols	$-\text{OH}$	hydroxy	ol
Amines	$-\text{NH}_2$	amino	amine
Aldehydes	$-\text{CHO}$	Oxo and formyl	al carbaldehyde
Ketones	$-\text{CO}-$	Oxo or keto	one
Carboxylic acid	$-\text{COOH}$	carboxy	oic acid carboxylic acid
Esters	$-\text{COOR}$	alkoxycarbonyl	oate carboxylate
Amides	$-\text{CONH}_2$	carbamoyl	Amide carboxamide
Acid halides	$-\text{COX}$	halocarbonyl	oyl halide carbonyl halide
Nitriles	$-\text{CN}$	cyano	Nitrile carbonitrile

Rules for multifunctional compounds:

- (i) Identify the main functional group using the priority order. The longest chain or **parent chain must contain the main functional group.**
- (ii) The main functional group is given the lowest possible number and is indicated by the secondary suffix. All other functional groups are indicated as prefixes.



3- Propylhex-1-ene (The longest chain has 6 carbons)



6-Hydroxy-3-methylhept-4-en-2-one

- (iii) If there are two different prefix functional groups at equivalent positions then the functional group which comes first in alphabetical order is given the lowest number.



1-Bromo-5-chloropentane.



1-Ethoxy-4-methoxybutane